

RDL-like sweep through inversion (msinv hull simulator)

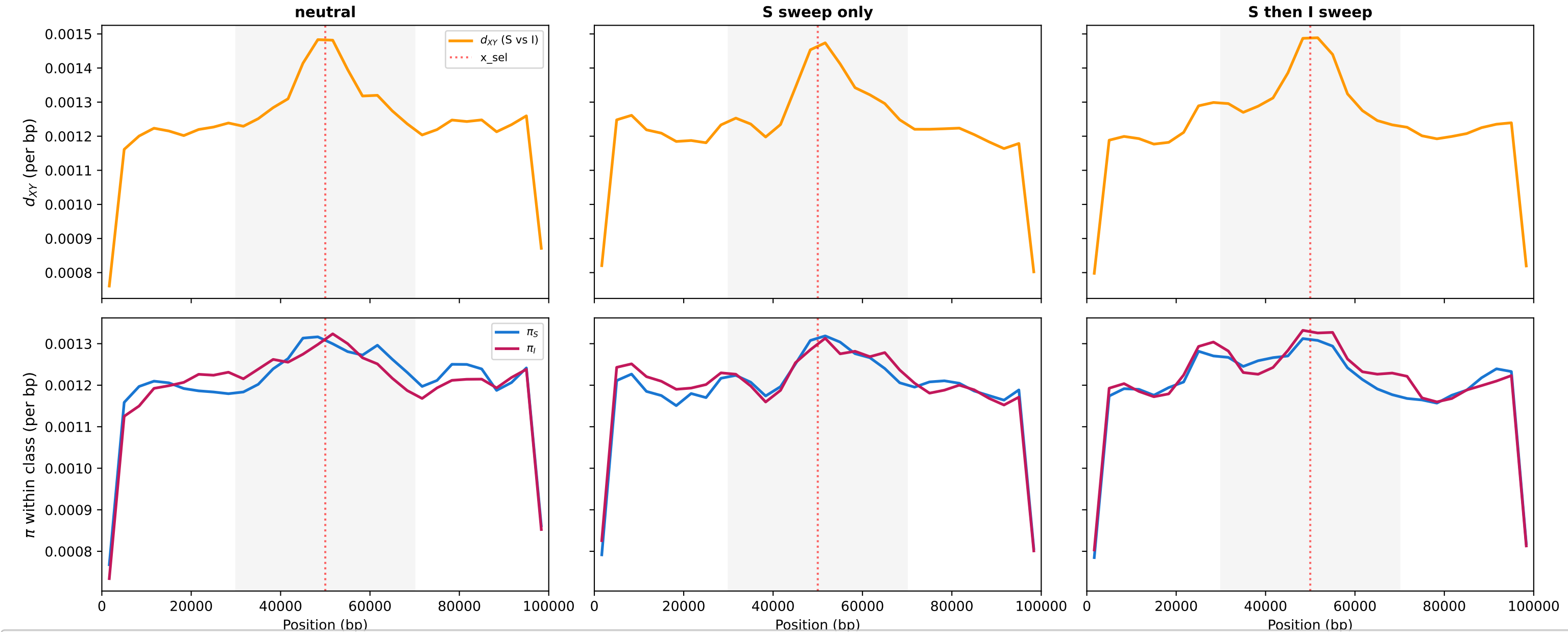


Figure. RDL (dieltrin resistance) sweep-through-inversion, modelled after Grau-Bové et al. (2020 MBE). Three scenarios for $n_S=8 + n_I=8$ haplotypes inside a single inversion. (Top) Cross-class d_{XY} (S vs I). (Bottom) Within-class π_S and π_I . Left: Neutral baseline — uniform elevation of d_{XY} inside the inversion. Centre: Sweep on S only ($t=700$ gen) — π_S collapses around x_{sel} while π_I is unaffected, producing haplotype asymmetry. Right: S sweep then I sweep (via gene flux at $t=500$ gen, then I sweep at $t=300$ gen) — both π_S and π_I collapse, matching the empirical RDL pattern. Parameters: $N_e=100,000$, $p_{inv}=0.5$, $t_{inv}=80,000$ gen, $\mu=3e-09$, $r=1e-08$, $L=100$ kb, 100 replicates.

Command: `pixi run -e all python examples/empirical_rdl_sweep.py`